SET 2 | SYSTEM ENGINEER

# Infosys
# Numerical Puzzle

## Set 2: 30 New Practice Questions

All-new puzzles with answers and step-by-step solutions.

| 30 | 4 | 100% |
|---|---|---|
| Questions | Options each | With solutions |

### WHAT'S INSIDE

**Number Series**
8 questions
Patterns, differences and alternating rules

**Grids & Magic Squares**
4 questions
Row and column rules, magic sums

**Sudoku & Cryptarithm**
4 questions
Digit logic and letter-to-number puzzles

**Weighing, Clock, Calendar**
4 questions
Classic reasoning with numbers

**Counting & Number Properties**
6 questions
Primes, zeros, remainders and digit counts

**Operators, Codes & Ages**
4 questions
Custom operators, letter codes, age problems

**HOW TO USE**

Give each question about 2 minutes, then check the answer and the solution below it.

# Infosys Numerical Puzzle

Set 2 | 30 New Questions

**1** Find the next term in the series: 7, 9, 13, 21, 37, ?

A) 63
B) 69
C) 65
D) 73

**Answer: B) 69**

The differences are 2, 4, 8, 16, so the next difference is 32. Hence 37 + 32 = 69.

**2** Find the next term in the series: 6, 11, 21, 41, 81, ?

A) 151
B) 161
C) 159
D) 163

**Answer: B) 161**

Each term is double the previous term minus 1. So 81 × 2 - 1 = 161.

**3** Find the missing term: 1, 8, 27, 64, 125, ?

A) 196
B) 206
C) 225
D) 216

**Answer: D) 216**

These are cubes: $1^3$, $2^3$, $3^3$, $4^3$, $5^3$. The next term is $6^3 = 216$.

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**4** Find the next term in the series: 2, 5, 10, 17, 26, 37, ?

A) 48
B) 50
C) 49
D) 52

**Answer: B) 50**

Each term is $n^2 + 1$ ($1^2 + 1$, $2^2 + 1$, ... $6^2 + 1 = 37$). The next term is $7^2 + 1 = 50$.

**5** Find the next term in the series: 120, 99, 80, 63, 48, ?

A) 31
B) 33
C) 36
D) 35

**Answer: D) 35**

The terms are $11^2 - 1$, $10^2 - 1$, $9^2 - 1$, $8^2 - 1$, $7^2 - 1$. The next term is $6^2 - 1 = 35$.

**6** Find the next term in the series: 3, 4, 7, 11, 18, 29, ?

A) 47
B) 45
C) 46
D) 48

**Answer: A) 47**

Each term is the sum of the previous two terms. So 18 + 29 = 47.

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**7** Find the next term in the series: 4, 6, 12, 14, 28, 30, ?

- A) 56
- B) 60
- C) 58
- D) 62

**Answer: B) 60**

The pattern alternates between + 2 and × 2. After 30 comes 30 × 2 = 60.

**8** Find the next term in the series: 2, 6, 7, 21, 22, 66, ?

- A) 67
- B) 63
- C) 69
- D) 72

**Answer: A) 67**

The pattern alternates between × 3 and + 1. After 66 comes 66 + 1 = 67.

**9** Every row follows the same rule. Find the missing number.

| | | |
|---|---|---|
| 2 | 3 | 11 |
| 3 | 4 | 19 |
| 4 | 5 | ? |

- A) 26
- B) 29
- C) 27
- D) 30

**Answer: B) 29**

In each row, third = first × second + first + second. So 4 × 5 + 4 + 5 = 29.

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**10** Every column follows the same rule. Find the missing number.

| | | |
|---|---|---|
| 3 | 6 | 7 |
| 1 | 5 | 4 |
| 8 | 31 | ? |

A) 45
B) 41
C) 43
D) 49

**Answer: A) 45**

In each column, (top)² - middle = bottom. So 7² - 4 = 49 - 4 = 45.

**11** This 3 × 3 grid is a magic square (all rows, columns and diagonals have the same sum). Find the number in the top-right cell.

| | | |
|---|---|---|
| 10 | 3 | ? |
| ? | 7 | 9 |
| 6 | ? | 4 |

A) 6
B) 7
C) 9
D) 8

**Answer: D) 8**

The completed square is 10 3 8 / 5 7 9 / 6 11 4 with a magic sum of 21. Top row: 21 - 10 - 3 = 8.

**12** Every row follows the same rule. Find the missing number.

| | | |
|---|---|---|
| 9 | 3 | 3 |
| 10 | 2 | 4 |
| 14 | 2 | ? |

A) 4
B) 5
C) 6
D) 8

**Answer: C) 6**

In each row, (first - second) ÷ 2 = third. So (14 - 2) ÷ 2 = 6.

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**13** In the cryptarithm SIX + SEVEN + SEVEN = TWENTY, each letter stands for a different digit. Find the value of W + E + N + T + Y.

A) 16
B) 17
C) 18
D) 20

**Answer: C) 18**

650 + 68782 + 68782 = 138214. So W = 3, E = 8, N = 2, T = 1 and Y = 4, and their sum is 18.

**14** AB and BA are two-digit numbers with AB + BA = 132 and A - B = 2. Find the value of A × B.

A) 30
B) 32
C) 35
D) 36

**Answer: C) 35**

AB + BA = 11(A + B) = 132, so A + B = 12. With A - B = 2, A = 7 and B = 5. Hence A × B = 35.

**15** Fill the 4 × 4 Sudoku so that every row, column and 2 × 2 box contains the digits 1 to 4. Find the sum of the eight missing digits.

| | | | |
|---|---|---|---|
| 3 | ? | 1 | ? |
| ? | 2 | 3 | ? |
| 4 | ? | ? | 1 |
| ? | 1 | 4 | ? |

A) 19
B) 20
C) 22
D) 21

**Answer: D) 21**

The completed rows are 3 4 1 2, 1 2 3 4, 4 3 2 1 and 2 1 4 3. The missing digits are 4, 2, 1, 4, 3, 2, 2 and 3, so the sum is 21.

# Infosys Numerical Puzzle



**16** Fill the 4 × 4 Sudoku so that every row, column and 2 × 2 box contains the digits 1 to 4. Which digit replaces the ? in the first row?

| | | | |
|---|---|---|---|
| 4 | ? | ? | 3 |
| ? | 3 | 4 | ? |
| 2 | ? | 3 | ? |
| ? | 1 | ? | 4 |

A) 2
B) 1
C) 3
D) 4

**Answer: A) 2**

The completed rows are 4 2 1 3, 1 3 4 2, 2 4 3 1 and 3 1 2 4. The first-row ? is 2.

**17** Among 27 identical-looking coins, one is slightly heavier. Using a balance scale, what is the minimum number of weighings that guarantees finding it?

A) 2
B) 4
C) 9
D) 3

**Answer: D) 3**

Each weighing narrows the heavy coin to a group one third the size: 27 to 9 to 3 to 1. So 3 weighings are enough.

**18** How many squares of all sizes are there in a 4 × 4 square grid?

A) 16
B) 24
C) 30
D) 36

**Answer: C) 30**

There are 16 squares of size 1, 9 of size 2, 4 of size 3 and 1 of size 4. Total = 16 + 9 + 4 + 1 = 30.

# Infosys Numerical Puzzle



**19** How many times do the hour hand and the minute hand of a clock overlap in one full day (24 hours)?

- A) 22
- B) 11
- C) 12
- D) 24

**Answer: A) 22**

The hands overlap 11 times in every 12 hours, so they overlap 22 times in 24 hours.

**20** If today is a Monday, which day of the week will it be 100 days from today?

- A) Tuesday
- B) Wednesday
- C) Thursday
- D) Friday

**Answer: B) Wednesday**

100 days = 14 weeks + 2 days. So the day is Monday + 2 days = Wednesday.

**21** How many times does the digit 7 appear when writing all the numbers from 1 to 100?

- A) 10
- B) 11
- C) 20
- D) 19

**Answer: C) 20**

The units digit is 7 in 10 numbers (7, 17, ... 97) and the tens digit is 7 in 10 numbers (70 to 79). Total = 20.

# Infosys Numerical Puzzle



**22** How many numbers from 1 to 60 are divisible by 3 or 5?

- A 24
- B 26
- C 32
- D 28

**Answer: D) 28**

Multiples of 3: 20. Multiples of 5: 12. Multiples of both (15): 4. So 20 + 12 - 4 = 28.

**23** How many zeros are there at the end of 25! (25 factorial)?

- A 4
- B 5
- C 7
- D 6

**Answer: D) 6**

Trailing zeros come from factors of 5: 25 ÷ 5 = 5 and 25 ÷ 25 = 1. So 5 + 1 = 6 zeros.

**24** How many prime numbers are there between 1 and 50?

- A 14
- B 15
- C 16
- D 17

**Answer: B) 15**

The primes are 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43 and 47, which is 15 primes.

# Infosys Numerical Puzzle



**25** What is the remainder when 2 raised to the power 50 is divided by 7?

A) 1
B) 2
C) 4
D) 3

**Answer: C) 4**

$2^3 = 8$ leaves remainder 1 when divided by 7. Since $50 = 3 \times 16 + 2$, the remainder is the same as for $2^2 = 4$.

**26** If a @ b = 2a + b, what is the value of (3 @ 4) @ 2?

A) 18
B) 22
C) 20
D) 24

**Answer: B) 22**

3 @ 4 = 2 × 3 + 4 = 10. Then 10 @ 2 = 2 × 10 + 2 = 22.

**27** If P # Q = $P^2 - Q$, what is the value of (4 # 3) # 5?

A) 164
B) 160
C) 162
D) 166

**Answer: A) 164**

4 # 3 = $4^2$ - 3 = 13. Then 13 # 5 = $13^2$ - 5 = 169 - 5 = 164.

# Infosys Numerical Puzzle



**28** If letters take their alphabet positions (A = 1, B = 2, ... Z = 26), what is the total value of the word ZERO?

- A) 64
- B) 60
- C) 62
- D) 66

**Answer: A) 64**

Z = 26, E = 5, R = 18 and O = 15. So the total is 26 + 5 + 18 + 15 = 64.

**29** The sum of the ages of a father and his son is 50 years. Five years ago, the father was 4 times as old as the son. What is the son's present age?

- A) 13
- B) 11
- C) 12
- D) 15

**Answer: A) 13**

Let the son be s, so the father is 50 - s. Then 45 - s = 4(s - 5), which gives 5s = 65 and s = 13.

**30** The product of two consecutive even numbers is 224. What is the sum of these two numbers?

- A) 28
- B) 32
- C) 30
- D) 34

**Answer: C) 30**

14 × 16 = 224, so the numbers are 14 and 16. Their sum is 30.

# Set 2 Answer Key



| Q1 | Q2 | Q3 | Q4 | Q5 |
|---|---|---|---|---|
| B | B | D | B | D |
| 69 | 161 | 216 | 50 | 35 |

| Q6 | Q7 | Q8 | Q9 | Q10 |
|---|---|---|---|---|
| A | B | A | B | A |
| 47 | 60 | 67 | 29 | 45 |

| Q11 | Q12 | Q13 | Q14 | Q15 |
|---|---|---|---|---|
| D | C | C | C | D |
| 8 | 6 | 18 | 35 | 21 |

| Q16 | Q17 | Q18 | Q19 | Q20 |
|---|---|---|---|---|
| A | D | C | A | B |
| 2 | 3 | 30 | 22 | Wednesday |

| Q21 | Q22 | Q23 | Q24 | Q25 |
|---|---|---|---|---|
| C | D | D | B | C |
| 20 | 28 | 6 | 15 | 4 |

| Q26 | Q27 | Q28 | Q29 | Q30 |
|---|---|---|---|---|
| B | A | A | A | C |
| 22 | 164 | 64 | 13 | 30 |